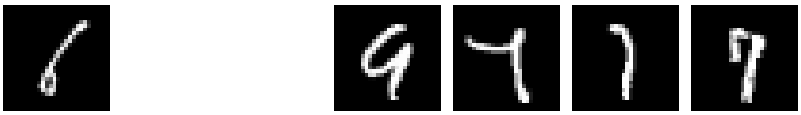

(a) The one test example that is incorrectly classified by all ten models. (Labeled as 6, predicted as 1.)

(b) Examples of rejected test examples for which models disagree on the predicted class.

Figure 4: Test examples that are misclassified or rejected by the ten-model ensemble.

We reran all the above experiments for all ϵ values from 0.01 to 0.20 to see how the joint robustness changes as the attacks get stronger. Figure 3(a) shows the results from the adversarially clustered seeds two-model ensemble; the results for the other ensembles show similar patterns and are deferred to Appendix A. For ϵ values up to 0.1, which is the value used for training the robust models, the ensemble model is able to certify more seeds compared to the single overall robust model. We also note that the models that are trained to be target-robust, rather than seed-robust, perform much worse. With even and odd target-robust models we were able to certify only 35.6% of test examples. We believe the reason for this is that the evaluation criteria of robustness is inherently biased against models that are trained to target-robust. Because, when evaluating, we always start from some test seed, and try to find an adversarial example from that seed – which is not what the target-robust models are explicitly trained to prevent.

Five-model Ensembles. Our joint certification framework can be extended to ensembles of any number of models. We trained the five models to be robust on modulo-5 seed digits. Ensembles of these models had better certified robustness than the best two-model ensembles. For example, with averaging composition 85.3% of test examples can be certified robust (compared to our previous best result of 78.1% with two models). Figure 3(b) shows how the number of certifiable test seeds drops with increasing ϵ, but worth noting is the large gap between any individual model's certifiable robustness and that for the average ensemble. We also trained model by adversarially clustering into 5-models – for digits (4, 9), (3, 5), (2,8), (0, 6) and (1, 7). For the clustered seed robust ensemble, the results were slightly worse (84.3%) than modulo-5 seeds robust model. One difference between the two-model and five-model ensembles is that in the latter, the unanimity and the majority frameworks are different. We found that independent certification does not really work for majority framework. We were able to certify almost no test seeds for the majority framework for five-model ensembles.

Ten-model Ensembles. Finally, we tried ensembles of ten models, each trained to be robust for a selected seed digit $(0, 1, 2, \ldots, 9)$. The certified robust rate of the 10-model ensemble trained to be seed robust was 85.6%. This is slightly higher than the 5-model ensemble (85.3%), but perhaps not worth the extra performance cost. It is notable, though, that the unanimity model reduces the normal test error for this ensemble to 0.1%. This means that out of 1000 test seeds, 853 were certified to be robust, 48 were correctly classified but could not be certified, 97 were rejected due to disagreement among the models, and 1 was incorrectly classified by all 10 models. Figure 4 shows the one test example where all models agree on a predicted class but it is not the given label (Figure 4(a), and selected typical rejected examples from the 97 tests where the models disagree (Figure 4(b)).

Summary. Figure 5 compares the robust certification rate for the two, five, and ten-model ensembles. Clustered seed robust models generally tend to perform well, although just random modulo seed robust models perform almost just as well.

One potential issue with any ensemble models is the possibility of false positives. In our case, the use of multiple models in the unanimity and majority frameworks also introduce the possibility of *rejecting* benign inputs. As the number of models in a unanimity ensemble increases, the rejection rate on normal inputs increases since if any one model disagrees the input is rejected. However, if the false rejection rate is reasonably low, then in many situations that may be an acceptable trade-off for higher adversarial robustness. The results in Table 2 are consistent with this, but show that even the ten-model unanimity ensemble has a rejection rate below 10%. For more challenging classification tasks, strict unanimity composition may not be an option if rejection rates become unacceptable, but could be replaced by relaxed notions (for example, considering a set of related classes as equivalent for agreement purposes, or allowing some small fraction of models to disagree).

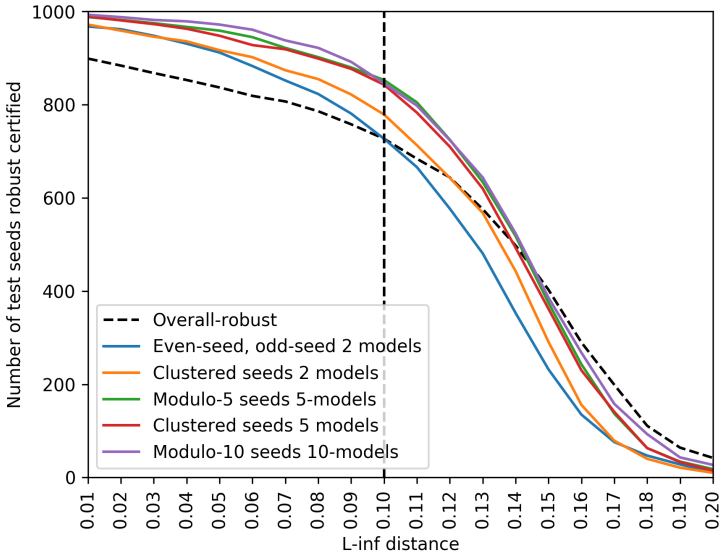

Figure 5: Number of test seeds certified to be jointly robust using ten models, five models, two models and single model for different ϵ values.

6 Conclusion

We extended robust certification models designed for single models to provide joint robustness guarantees for ensembles of models. Our novel joint-model formulation technique can be used to extend certification frameworks to provide certifiable robustness guarantees that are substantially stronger than what can be obtained using the verification techniques independently. Furthermore, we have shown that cost-sensitive robustness training with diverse cost matrices can produce models that are diverse with respect to joint robustness goals. The results from our experiments suggest that ensembles of models can be useful for increasing the robustness of models against adversarial examples. These is a vast space of possible ways to train models to be diverse, and ways to use multiple models in an ensemble, that may lead to even more robustness. As we noted in our motivation, however, without efforts to certify joint robustness, or to ensure that models in an ensemble are diverse in their vulnerability regions, the apparent effectiveness of an ensemble may be misleading. Although the methods we have used cannot yet scale beyond tiny models, our results provide encouragement that ensembles can be constructed that provide strong robustness against even the most sophisticated adversaries.

Availability

Open source code for our implementation and for reproducing our experiments is available at: https://github.com/jonas-maj/ensemble-adversarial-robustness.

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